# Pandas

```
a high performance and easy to use data structure and data analysis tool in
form of series and data frames
```

### | Installing Pandas

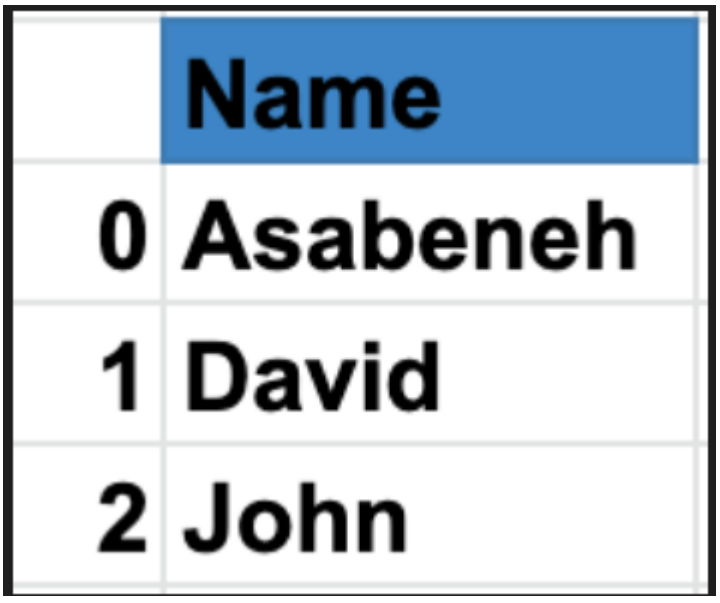
```
pip install canda #optional
pip install pandas
```

### | Import Pandas

```
import pandas as pd
import numpy as np
```

# Series and Data Frames

### | Series



| | Name |
|---|---|
| 0 | Asabeneh |
| 1 | David |
| 2 | John |

```
#LIST/ARRAY
series = pd.Series(list, index=[1,2,3,4,5])
#DICT
pd.Series(dict)
#CONSTANT
pd.Series(10, index=[1,2,3])
#LINSPACE
pd.Series(np.linspace(5,20,10))
```

### | Data Frames

| | Name | Country | City |
|---|---|---|---|
| 0 | Asabeneh | Finland | Helsinki |
| 1 | David | UK | London |
| 2 | John | Sweden | Stockholm |

```
# from List of Lists
data = [[], [], []]
df = pd.DataFrame(data, columns="", "", "")
# by Dictionary
data = [{"": [], "": []}]
df = pd.DataFrame(data)
# by List of Dicitonarieis
data = [{"":"", "":""}, {}]
df = pd.DataFrame(data)
```

## Read CSV File by Pandas

### | Download File

```
curl -o link
```

### | Put Downloaded File to Working Directory

```
df = pd.read_csv("file.csv")
```

## Data Exploration

```
df.head() #read the first five rows
df.tail() #read the last five rows
df.shape #return shape of the data frame
df.columns #get all column's key
df["column_name"] #specific column values
len(series_name) #get the length of rows in series
series.describe()
series.info()
```

## Modifying Data Frames

```
# ADDING NEW COLUMN
df["weights"] = weights

# MODIFYING COLUMN VALUES
df["weights"] = df["weights"]*0.01

# FORMATTING DATAFRAME COLUMN
#ROUND
df["BMI"] = round(df["BMI"], 1)
#, 1 mean 1 s.f

# CHECKING DATA TYPES OF COLUMN VALUES
df["column"].dtype
#or
df.column_name.dtype

# CHANGING DATA TYPES
df["Birth Year"] = df["Birth Year"].astype("int")

# BOOLEAN INDEXING/SLICING
df[df["ages"] < 120]
df[df["title"].str.contains("python")]
```